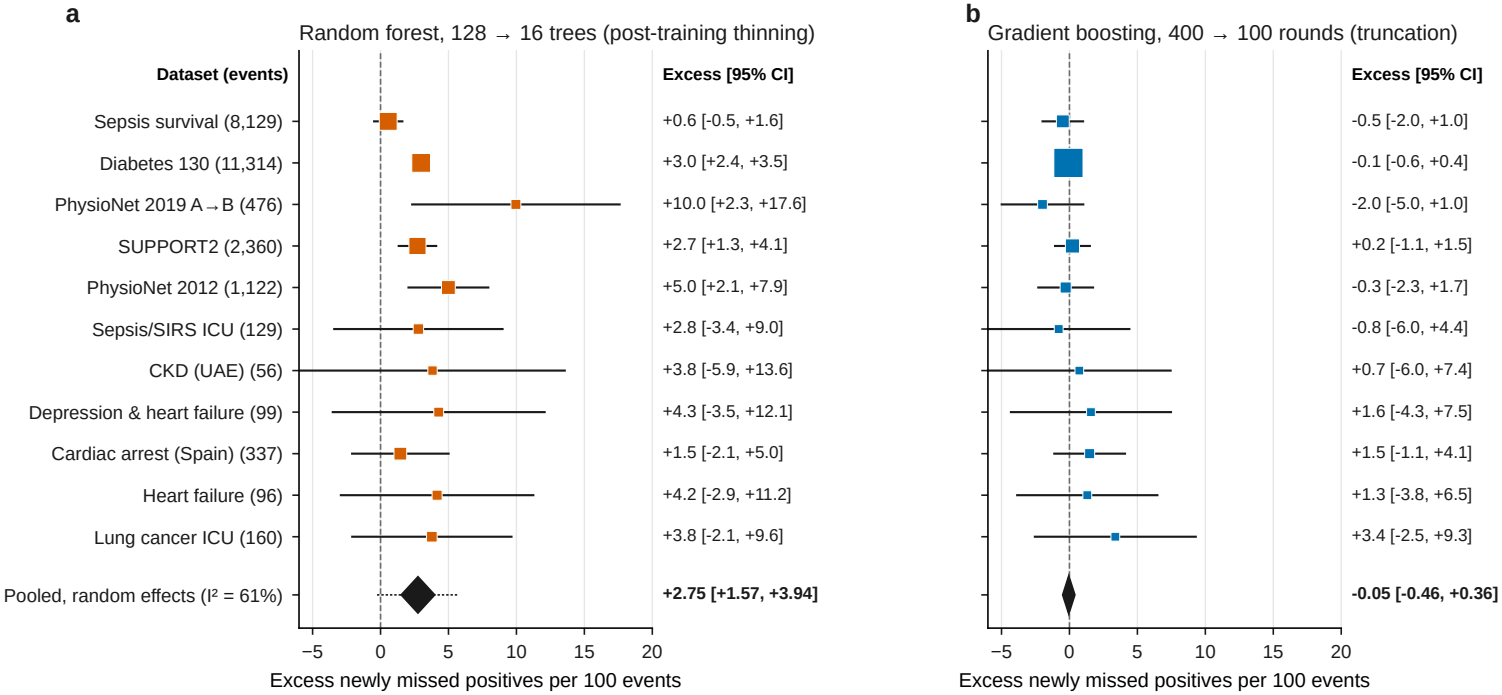
a
Random forest, 128 → 16 trees (post-training thinning)
Dataset (events)
Excess [95% CI]
Sepsis survival (8,129)
+0.6 [-0.5, +1.6]
Diabetes 130 (11,314)
+3.0 [+2.4, +3.5]
PhysioNet 2019 A→B (476)
+10.0 [+2.3, +17.6]
SUPPORT2 (2,360)
+2.7 [+1.3, +4.1]
PhysioNet 2012 (1,122)
+5.0 [+2.1, +7.9]
Sepsis/SIRS ICU (129)
+2.8 [-3.4, +9.0]
CKD (UAE) (56)
+3.8 [-5.9, +13.6]
Depression & heart failure (99)
+4.3 [-3.5, +12.1]
Cardiac arrest (Spain) (337)
+1.5 [-2.1, +5.0]
Heart failure (96)
+4.2 [-2.9, +11.2]
Lung cancer ICU (160)
+3.8 [-2.1, +9.6]
Pooled, random effects (I² = 61%)
+2.75 [+1.57, +3.94]
−5
0
5
10
15
20
Excess newly missed positives per 100 events
b
Gradient boosting, 400 → 100 rounds (truncation)
Excess [95% CI]
-0.5 [-2.0, +1.0]
-0.1 [-0.6, +0.4]
-2.0 [-5.0, +1.0]
+0.2 [-1.1, +1.5]
-0.3 [-2.3, +1.7]
-0.8 [-6.0, +4.4]
+0.7 [-6.0, +7.4]
+1.6 [-4.3, +7.5]
+1.5 [-1.1, +4.1]
+1.3 [-3.8, +6.5]
+3.4 [-2.5, +9.3]
-0.05 [-0.46, +0.36]
−5
0
5
10
15
20
Excess newly missed positives per 100 events